

Anastasiia Mokhonko

DATA SCIENCE & AI STUDENT

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“Aspiring Data Scientist and AI Specialist passionate about digital innovation, machine learning, and solving real-world problems.”

Summary

Machine learning engineer with hands-on experience across the full AI lifecycle — from research and model development to production deployment. Currently interning as an AI Product Engineer at Vanderlande (CDAO), building an automated code generation tool on Databricks and Azure that targets a 50–60% reduction in engineering time. Previously developed custom deep learning architectures for CT bone segmentation at a confidential medical technology company, achieving clinical-grade accuracy with 90% fewer parameters than the state-of-the-art baseline. My broader project portfolio spans NLP, computer vision, and MLOps across clients including Content Intelligence Agency, NPEC, NAC Breda, and ANWB. I apply regulatory frameworks including GDPR, EU AI Act, and EU MDR compliance as standard practice.

Experience

Vanderlande – Central Data & Analytics Organization (CDAO)

Veghel, NL

AI PRODUCT ENGINEER INTERN

Feb 2026 – Jul 2026

- Building an end-to-end automated code generation tool eliminating 7–10 hours of manual work per Data Hub implementation, targeting 50–60% reduction in engineering time across Vanderlande’s enterprise data lake platform (Databricks, Azure).
- Implemented a production-grade Python pipeline parsing structured Excel/CSV mapping documents to auto-generate four standardized output files per Data Hub, validated across 11+ hubs spanning Finance, HR, Supply, and Technology domains.
- Integrated Meta Llama 3.3 70B via Databricks MLflow Deployments, combining LLM assistance with template-based generation; benchmarked against M365 Copilot achieving ~98% vs ~80% accuracy.
- Deployed an interactive Gradio UI via Databricks Apps for non-technical engineers; maintained a 59-test Python unit suite ensuring cross-domain reliability and regression coverage.
- Operating within a SAFe Agile Release Train, contributing to sprint planning, backlog refinement, and stakeholder demos.

Bone Segmentation for Surgical Planning – Confidential Medical Technology Company

Breda, NL

AI RESEARCHER, DEEP LEARNING ENGINEER

Sep 2025 – Jan 2026

- Developed custom ResU-Net deep learning architectures for automated radius and ulna bone segmentation from CT scans, achieving 86.9% test Dice score — within 3.1 pp of the nnU-Net state-of-the-art benchmark.
- Reduced model complexity by 90% (3.2M vs 31M parameters) and training time by 89% (7 hours vs 63 hours) while maintaining clinical-grade accuracy above the 85% acceptability threshold.
- Built and validated an 8-step preprocessing pipeline (voxel standardization, intensity normalization, patch extraction, HDF5 batching) across 162 CT cases from Radboud University Medical Center.
- Conducted 150+ training runs comparing U-Net, ResU-Net, V-Net, and nnU-Net architectures; delivered regulatory and ethical compliance framework for EU MDR Class IIa medical device deployment.

Emotion Detection NLP Pipeline – Content Intelligence Agency (Phase 2)

Breda, NL

MLOPS ENGINEER, DEVELOPER

May 2025 – Jun 2025

- Deployed production-ready emotion detection system classifying 6 emotions in French video/audio content with 44.7 requests/second capacity.
- Built Azure ML pipelines with automated training, retraining, and deployment using Docker containers and CI/CD integration.
- Developed secure FastAPI endpoints and Streamlit interface for multi-user access with monitoring and logging.
- Implemented blue-green deployment strategy with model versioning and rollback capabilities for zero-downtime updates.

Emotion Detection from Video – Content Intelligence Agency (Phase 1)

Breda, NL

DEVELOPER, NLP ENGINEER

Feb 2025 – Apr 2025

- Developed NLP pipeline for emotion analysis in French reality TV including audio transcription, translation, and classification.
- Implemented and compared Naive Bayes, Logistic Regression, LSTM, BERT, and CamemBERT models for emotion detection.
- Applied XAI techniques (LIME, SHAP) for model interpretability; delivered research documentation and model card to client.

Root Segmentation & Robotic Control – NPEC Netherlands Plant Eco-phenotyping Centre

Breda, NL

DEVELOPER, COMPUTER VISION ENGINEER

Nov 2024 – Dec 2024

- Built computer vision pipeline using U-Net for plant root segmentation and graph theory (NetworkX, Dijkstra) for measurement.
- Developed PID and Reinforcement Learning controllers (Stable Baselines 3) for Opentrons OT-2 robotic microbial application.
- Achieved 99% time reduction in root analysis workflow; delivered high-accuracy solution to research client.

Cybersecurity Awareness in Small Hotels – Policy Research for DigiWerkplaats

Breda, NL

ANALYTICS TRANSLATOR, TEAM LEAD, RESEARCHER

Sep 2024 – Oct 2024

- Led research team of 5 investigating employee cybersecurity awareness impact on hotel security using mixed-methods approach.
- Designed and analyzed surveys (Qualtrics) and interviews; ensured GDPR compliance and research ethics approval.
- Found moderate positive correlation (r=0.456) between training frequency and security adherence; recommended quarterly training.

ANWB Safe Driving Application – ANWB Dutch Automobile Association

Breda, NL

DEVELOPER, DATA ANALYST

May 2024 – Jun 2024

- Developed Streamlit application analyzing driver behavior and road safety using ANWB datasets and time series analysis.
- Implemented incident classification models with EU AI Act compliance and Agile project management methodology.

Farmer’s Buddy: Potato Blight Detection AI

Breda, NL

DEVELOPER, DATA SCIENTIST

Feb 2024 – Apr 2024

- Built AI-driven mobile application for potato blight detection using MobileNetV2 transfer learning, achieving 95.3% accuracy.
- Applied Grad-CAM for explainable AI and conducted think-aloud usability studies to optimize user experience.

Goal Prediction Analytics – NAC Breda Football Club

Breda, NL

DATA ANALYST, MACHINE LEARNING ENGINEER

Nov 2023 – Jan 2024

- Developed machine learning model analyzing 16,535 players to predict goal-scoring potential with 95%+ accuracy.
- Delivered GDPR-compliant recruitment recommendations using linear regression and correlation analysis.

Human Trafficking Data Analysis Dashboard – UN SDG 16

Breda, NL

DEVELOPER, DATA ANALYST

Sep 2023 – Nov 2023

- Designed interactive Power BI dashboard analyzing global trafficking patterns by age, gender, exploitation type, and region.
- Calculated temporal correlation (r=0.68) and delivered policy recommendations for prevention strategies supporting SDG 16.

Education

Breda University of Applied Sciences

Breda, Netherlands

BACHELOR OF APPLIED DATA SCIENCE & ARTIFICIAL INTELLIGENCE

Sep 2023 – Present

ITSTEP Academy

Kyiv, Ukraine

PYTHON DEVELOPER COURSE

2022 – 2023

- Completed 6-month intensive course covering Python, OOP, algorithms, SQL/NoSQL, Django, Git, and cloud technologies.

Extracurricular Activity

Degree Programme Committee (DPC) – Data Science & AI Programme

Breda University of Applied Sciences, Breda, NL

STUDENT REPRESENTATIVE

2024 – 2026

- Selected as one of three student representatives to collaborate with faculty on curriculum quality, course delivery, and student experience.
- Act as liaison between student body and programme management, representing student perspectives in policy and assessment decisions.

Nakormim Bali – Orphanage Support Organization

Bali, Indonesia

VOLUNTEER & MONTHLY DONOR

2022 – Present

- Support post-COVID-19 relief for orphanages through monthly donations, educational sessions, and coordinated activities for children.

Technical Skills

Programming	Python, SQL, HTML/CSS, LaTeX
ML/AI Frameworks	PyTorch, TensorFlow, Scikit-learn, Transformers (HuggingFace), BERT, CamemBERT, U-Net,
MLOps & Cloud	Azure ML, Azure DevOps, Databricks (Unity Catalog), MLflow, Docker, CI/CD, Git, FastAPI
Data & Visualization	Power BI, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Gradio, Apache PySpark, SQL (PostgreSQL, MySQL)
Specializations	NLP, Computer Vision, Deep Learning, LLM Integration, Time Series Analysis, XAI (LIME, SHAP)
Tools & Platforms	VSCode, Jupyter, Git/GitHub, Figma, Trello, Notion

Languages

Native	Russian, Ukrainian
Professional	English (C1 - Advanced)
Intermediate	French (B2 - Upper Intermediate)